

Research Notes Template - Team Name

Date:

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Researcher(s):

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Objective of research:

Research the guidelines of the MATE ROV project and make connections to the C&C responsibilities

Notes:

To avoid the deduction of points for the project:

- All required documentation must be submitted by given deadlines
- Submissions must be within file limits
- Submissions have proper naming conventions that must be followed
- PDFs only
- SID must show a fuse
- Fuse must use ANDI, NEMA, or IEC symbol
- Full load amps value and fuse selected must be shown on SID
- Non-ROV devices must be shown on SID
- Non-ROV devices used must be properly documented
- Companies must use fluid power

Must complete onsite safety inspection before vehicle enters the water

After the vehicle passes safety inspection, the company get a blue "PASSED" card from the safety inspector

The blue "PASSED" card must be given to product demonstration coordinator before vehicle can enter water

Companies can prepare for the safety inspection by using the safety inspection rubric
ALL items used on the ROV go through safety inspection

The ROV must be ONE body, no floating parts or separate vehicles are allowed

The ROV electronic housings need to be capable of functioning up to 7 meters under the water

There is a weight limit of 35 kilograms for all ROVs, as C&C we will have to ensure the weights of all our parts leaves enough room and weight for the rest of the teams to operate without issues

ROVs need to be capable of operating up to 5.5 meters under the water

All ROV underwater demonstrations will be no more than 8 meters away from the side of the pool

All propellers must be shrouded and have thruster guards

ROVs must be able to run off of either 48-volts or 12-volts, the C&C team must ensure that all the electronic parts (microcontrollers, etc.) are able to operate using the chosen power supply

Companies using 12 volts MUST use a proper blade fuse

AAA, AA, C, D, and 9V alkaline batteries are allowed

NiMH and AGM batteries also are allowed

Any ROV system disconnected from surface supply must stop functioning within 5 seconds, as C&C we'll have to make sure that all of our electrical components follow this rule

All pneumatic devices must be rated for at least 2.5 times the maximum supply pressure

Sources:

2025 Mate Competition Manual Page covers. (2024, November).

<https://20693798.fs1.hubspotusercontent->

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